

MATERIALS SCIENCE

An overbend strategy to manifold enhance the designed elastic bendability of flexible electronics

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Elastic bendability is crucial for flexible electronics to achieve conformal contact and enable widespread applications. However, reducing the thickness of the structure is almost the only strategy to enhance the elastic bendability of inorganic flexible electronics. This study proposes an overbend strategy, whose operation is beyond the designed elasticity of the fabricated structure. Mechanical theories, finite element analysis, and experiments verified that, with multilayer stacks and interconnects with various geometric designs as examples, the overbend strategy can enhance the designed elastic bendability to twice for the material of the ideal elastoplastic/kinematic hardening constitutive relationship and to more than twice for the mixed hardening/isotropic hardening constitutive relationship. The mechanism is extending the elastic strain range at the critical section to twice or more times the original value by the evolution of the elastoplastic constitutive relationship. Notably, the overbend strategy is virtually costless, combinative with other strategies and widely applicable for inorganic flexible electronics.

INTRODUCTION

The unique elastic bendability and stretchability of flexible electronics enable them to wrap conformally around complex surfaces, which facilitates widespread applications in fields such as health monitoring (1–7), medical treatment (8–10), intelligent robotics (11–16), etc. The elastic bendability of inorganic flexible electronics is attributed to the small thickness of the single-layer metal foil or multilayer stacks (Fig. 1, A and B) (17–20). On the other hand, the stretchability of partial structures is also achieved through the out-of-plane bending of the micro-interconnects (17). Compared to the plastic and failure behaviors, the elastic behaviors have attracted most of the research attention because of performance requirements of practical applications. For precise flexible electronics, such as flexible sensors (21), hemispherical electronic eyes (22), etc., reversibility, repeatability, and long-term stability in a certain deformation range are required for flexible structures, all of which correspond to the elastic behaviors. In addition, the long fatigue life could be achieved if the deformation remains within the elastic range (23), which is also a critical design objective for many flexible electronics. Therefore, the strategy of enhancing the elastic bendability of macro- and microstructures is fundamentally critical for the design of flexible electronics.

Although thousands of research papers in the field of inorganic flexible electronics have adopted bendable structures that are accomplished by mechanical design, most of them focused on only two types of forms with similar mechanical principles. (i) Single-layer metal foil (24). The elastic bendability is inversely proportional to the metal thickness according to the analysis of mechanics of materials, as shown in the upper inset of Fig. 1B (17). The reduction of the thickness is the commonly-used strategy to enhance the elastic bendability. (ii) Multilayer stacks (25–33). The stack generally consists of conductive metallic layers and nonconductive polymer layers,

such as polyimide. As shown in the lower inset of Fig. 1B, the elastic bendability is inversely proportional to the thickness of the entire layer. Consequently, the strategy to enhance the elastic bendability of this type of structure is to reduce the thickness of multilayer stacks.

Here, we propose an overbend strategy with the operation beyond the designed elastic bendability of the structure after the fabrication. The strategy uses the evolution of the elastoplastic constitutive relationship to achieve twice or more times the elastic strain range in the crucial section of the structure. Thus, it can effectively enhance the designed elastic bendability to twice for the material of the ideal elastoplastic/kinematic hardening constitutive relationship and to more than twice for the mixed hardening/isotropic hardening constitutive relationship, which is useful to multilayer stacks and interconnects with various geometric designs. The overbend strategy can be extended to the general overload strategy for flexible electronics with complex deformation modes including coupling bending, stretching, twisting, etc., by which the elastic range can be manifold enhanced.

RESULTS

Demonstration of the overbend strategy

Figure 1C demonstrates the operations of the overbend strategy to enhance elastic bendability via the multilayer stack with $t_{\text{metal}} = 70 \mu\text{m}$ for the upper and lower metal layers, $t_{\text{polymer}} = 260 \mu\text{m}$ for the polymer interlayer, and $L \times w = 1 \text{ mm} \times 0.5 \text{ mm}$ for the entire structure. The left column presents the detailed operation of the overbend strategy, where both loading and unloading are in the pure bending state. The middle and right columns display the maximum principal strain contour and equivalent plastic strain contour for each step in the finite element analysis (FEA). For simplicity, the ideal elastoplastic and linear elastic models are adopted for the mechanical constitutive relationships of the metal layers and polymer layer, respectively. To be specific, Young's modulus $E_{\text{metal}} = 124 \text{ GPa}$, $\sigma_s = 65 \text{ MPa}$, and $\nu_{\text{metal}} = 0.33$ are used for copper, and $E_{\text{polymer}} = 20 \text{ GPa}$ and $\nu_{\text{polymer}} = 0.38$ are used for glass resin (34).

In state ①, the entire structure is stress/strain-free. In state ②, the maximum stress reaches the yield stress σ_s at the outermost of the

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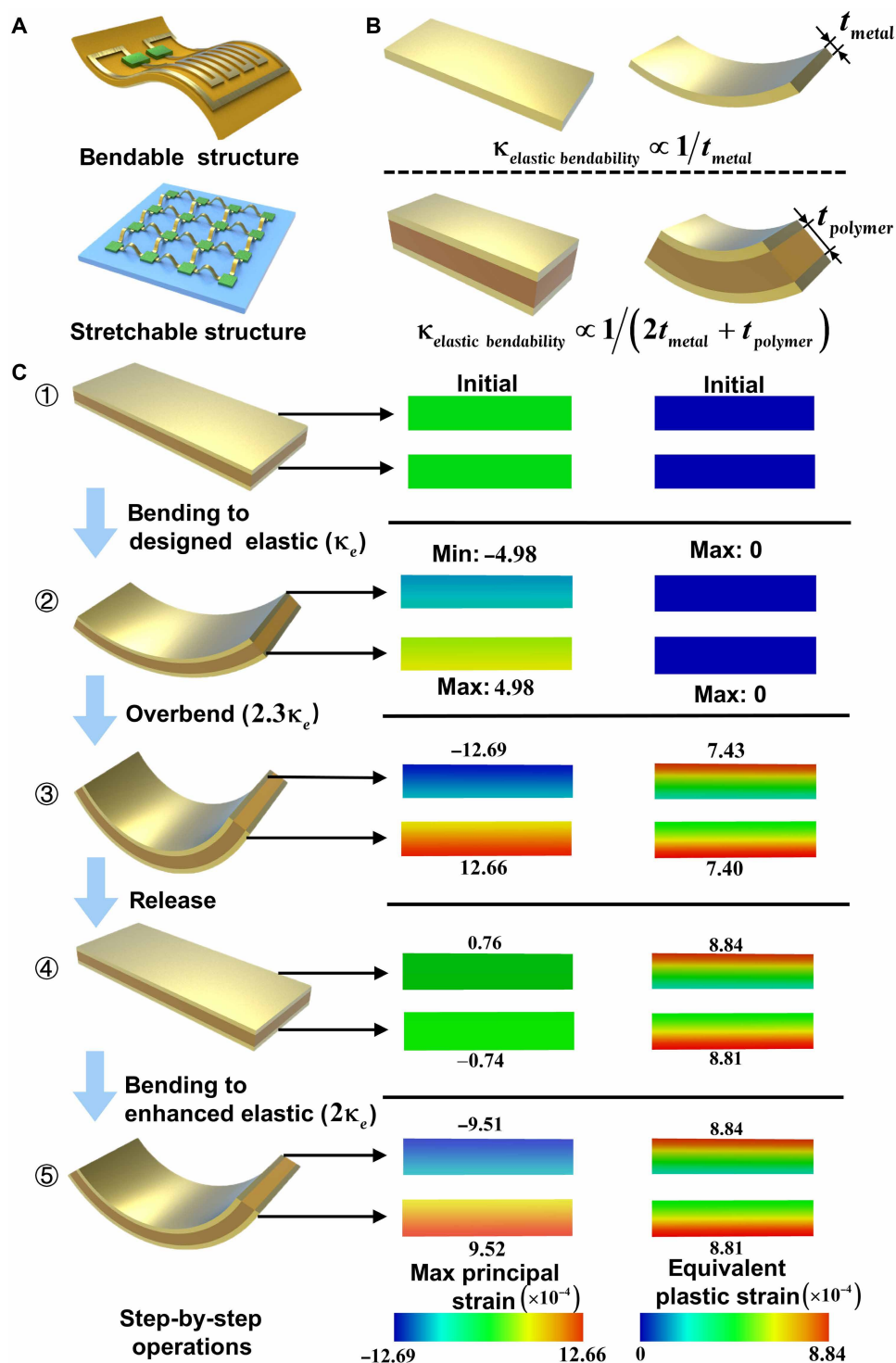


Fig. 1. Operations of the overbend strategy. (A) Examples of bendable and stretchable structures. (B) Relationship between the elastic bendability and the thickness of different structures. (C) Operations of the overbend strategy. The left column shows the step-by-step operations: ① to ②: bending to designed elastic bendability; ② to ③: bending beyond designed elastic bendability; ③ to ④: releasing the structure; ④ to ⑤: bending to enhanced elastic bendability. The middle and right columns show the contours of the FEA for the maximum principal strain and the equivalent plastic strain of the metal layer corresponding to each step, respectively.

metal layer, indicating that the structure reaches the designed elastic bendability κ_e . During bending from state ① to state ②, the deformation of the entire structure remains purely elastic without plasticity. With the increase in bending, the plastic behavior occurs in the metal layers from state ② to state ③. The outermost portion of the metal layer yields first, followed by the gradual expansion of the plastic region inward and the increase in the equivalent plastic strain. The curvature of the first bending increases until $2.3\kappa_e$, i.e., 2.3 times the designed elastic bendability, which is a certain value beyond $2\kappa_e$ as an example. After releasing the overbend, the polymer layer nearly returns to its original state, bringing the metal layers back to their initial condition, as shown in state ④. During this process, the plastic region on the cross section undergoes elastic reverse unloading and plastic reverse unloading sequentially, resulting in a more complex stress distribution on the cross section. The principal strain decreases, while the equivalent plastic strain increases. From state ① to state ④, the overbend strategy has been executed.

Twice the designed elastic bendability is applied from state ④ to state ③, while no additional equivalent plastic strain occurs in the entire stack during this process (fig. S1). The increase in the maximum principal strain is purely elastic and is twice the increment from state ① to state ②. Here, the enhanced elastic bendability is defined as the maximum value of the second bending, which does not result in any additional plasticity. In summary, after completing the overbend strategy from state ① to state ④, the enhanced elastic bendability can achieve twice the designed elastic bendability (see details in Supplementary Text 1).

Mechanism analysis and experimental verification via the multilayer stack

To quantitatively describe the mechanism of the overbend strategy, the multilayer stack is used as an example and the materials are the same as above. On the basis of the beam theory with plane-section assumption (see details in Supplementary Text 2), the designed elastic bendability can be calculated as

$$\kappa_e = \frac{\sigma_s}{E_{\text{metal}}(t_{\text{metal}} + t_{\text{polymer}}/2)} \quad (1)$$

During the bending from state ① to state ② (Fig. 2A), the deformation remains purely elastic, with linear stress distribution across the cross sections of each layer. Suppose that the designed elastic bendability is not exceeded in the first bending, the elastic bendability of the stack is not enhanced, as indicated by the horizontal line on the left side of Fig. 2B. As the deformation increases during the first bending, the plastic behavior occurs in the metal layers, as shown in state ③ $\kappa = \kappa_{\text{overbend}}$. The critical issue during the reverse unloading (from state ③ to state ④ $\kappa = 0$) is whether plastic behavior occurs again. This can be analyzed by the stress state during the reverse unloading process. Assuming that only elastic reverse unloading occurs, without any plastic reverse unloading, the maximum stress change in the metal can reach $2\sigma_s$. By substituting σ_s with $2\sigma_s$ into Eq. 1, the range of overbend curvature can be determined, denoted as $\kappa_{\text{overbend}} \leq 2\kappa_{\text{designed elastic}}$. Vice versa, from state ④ to state ⑤, when the second bending reaches the first bending value κ_{overbend} , the entire stack does not generate any plastic behavior again and the stress variation is linear in each layer, as shown in the second row of subfigures in Fig. 2A. Therefore, the enhanced elastic bendability can be obtained as

$$\kappa_{\text{enhanced elastic}} = \kappa_{\text{overbend}}, \text{ for } \kappa_{\text{designed elastic}} \leq \kappa_{\text{overbend}} \leq 2\kappa_{\text{designed elastic}} \quad (2)$$

as indicated by the diagonal line segment in the middle of Fig. 2B.

In the case where $\kappa_{\text{overbend}} > 2\kappa_{\text{designed elastic}}$, additional plastic behavior occurs during the reverse unloading from state ③ to state ④. However, during the second bending, the stress change in the metal layer still does not exceed $2\sigma_s$, as shown in the third row of subfigures in Fig. 2A. Therefore, after the execution of the overbend, the enhanced elastic bendability of the multilayer stack does not exceed twice the designed elastic bendability, which can be obtained as

$$\kappa_{\text{enhanced elastic}} = 2\kappa_{\text{designed elastic}}, \text{ for } \kappa_{\text{overbend}} \geq 2\kappa_{\text{designed elastic}} \quad (3)$$

as shown by the horizontal line segment on the right side of Fig. 2B. On the basis of these results, the first bending has an optimal value: $\kappa_{\text{overbend}} = 2\kappa_{\text{designed elastic}}$. At this value, the stack can achieve twice the designed elastic bendability by the overbend strategy without causing unnecessary plasticity. The analytical results presented above are derived through logical deduction. Readers interested in the detailed mechanical evolution process may refer to the elastoplastic theory of the beam, which has been well developed by L. Euler, D. Bernoulli, Timoshenko (35), Levinson (36), and others (37–48) (see details in Supplementary Text 3). In addition, the residual normal stress generated during the manufacture does not affect the twice enhancement for the elastic bendability (see details in Supplementary Text 4).

By focusing on the evolution of the constitutive relationship in the outermost metal layer where the stress changes the most and yields first, the intrinsic mechanism of the overbend strategy can be better understood. As shown in Fig. 2C, the green, red, and blue arrows represent the stress path during the first bending, reverse unloading, and the second bending, respectively. It can be observed that when the reverse unloading stress does not yield ($1.5\kappa_e$), the stress curve during the second bending coincides with the stress curve during the reverse unloading. In this case, the enhanced elastic bendability is the same as the first bending value. When the reverse unloading stress yields ($3\kappa_e$), the stress curve during the second bending no longer coincides with the stress curve during the reverse unloading, but the maximum stress variation is still limited to $2\sigma_s$. In summary, the essence of the overbend strategy is to shift the constitutive relationship of the critical regions from $0 \sim \sigma_s$ to $-\sigma_s \sim \sigma_s$ for the ideal elastoplastic model. As a result, both the elastic strain range in the constitutive relationship and the elastic bendability of the multilayer stack become twice as large as their original values. It is important to note that the theoretical result is based on the small deformation assumption, which explains the mechanical mechanism without losing the essence. Finite element analyses are performed for both small deformation and finite deformation assumptions (fig. S2), proving that the influence of geometric nonlinearity is negligible and can be disregarded. Besides the glass resin for the substrate, the effectiveness of the overbend strategy is also verified by FEA for the substrate of polyimide, polydimethylsiloxane, and polyethylene terephthalate (fig. S3).

To form the pure bending state, the four-point bending experiment is conducted in this study. In the central span, the bending moment is uniform according to the mechanical analysis, as shown in fig. S4, while it is linearly distributed in the bilateral stacks. The maximum bending moment of the bilateral stacks is equal to that of the central span. The above theoretical results of the enhanced

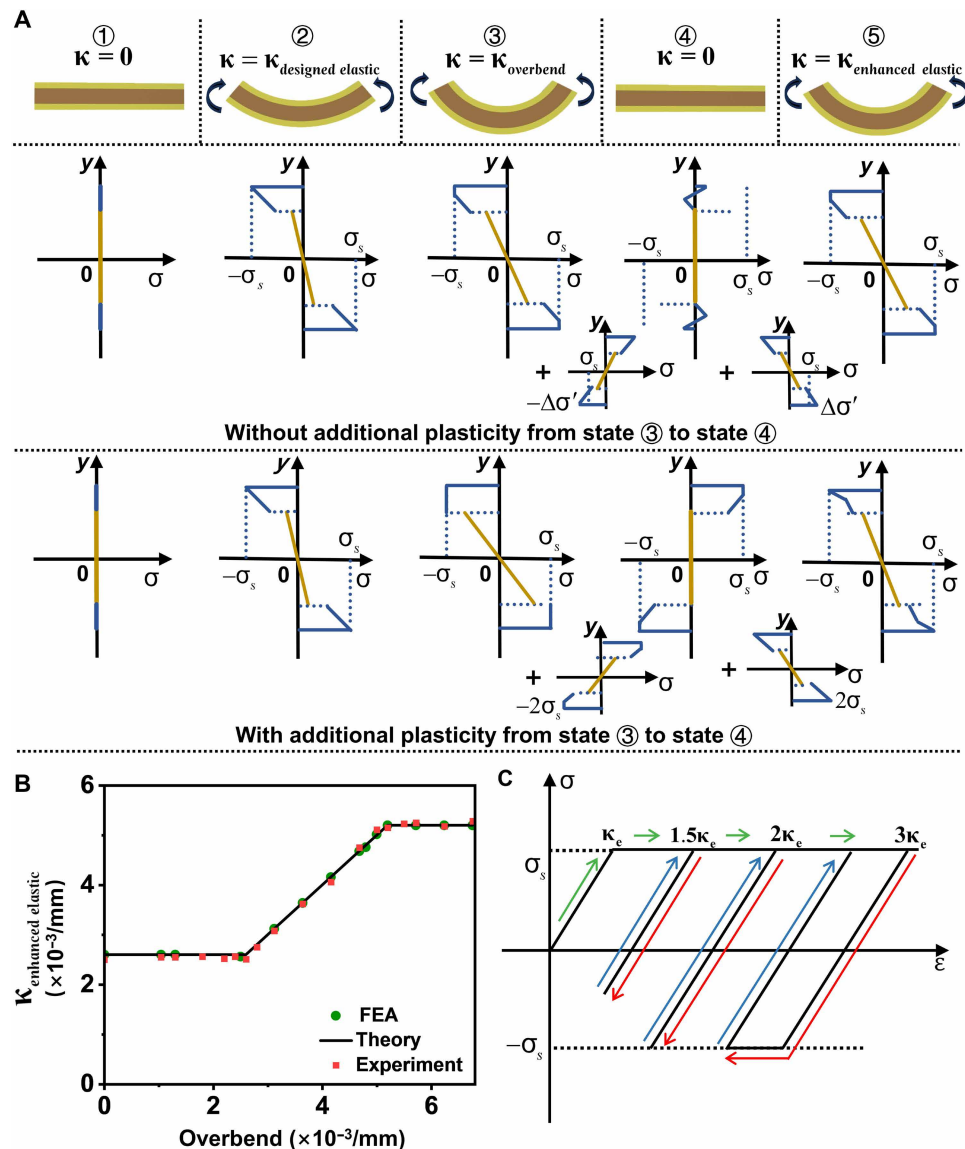


Fig. 2. Mechanical analysis of the overbend strategy. (A) First row: step-by-step overbend strategy operations with the multilayer stack. Second row: stress distribution in the cross section in each process if no additional plasticity occurs from state ③ to state ④. Third row: stress distribution in the cross section in each process if additional plasticity occurs from state ③ to state ④. (B) Enhanced elastic bendability as a function of the maximum of the first bending, including the results of the experiments, FEA, and theory. (C) Evolution of the ideal elastoplastic constitutive relationship for different values of overbend operation.

elastic bendability can be verified by the force-deflection curves of loading-unloading cycles with elastoplastic behaviors. The ideal elastoplastic constitutive relationship of copper (Fig. 3A) is obtained by the tensile test (fig. S5), and the geometric parameters of the stack for the four-point bending experiment are given in figs. S6 and S7. The four-point bending experiment (Fig. 3B) includes four steps: first bending, first reverse unloading, second bending, and second reverse unloading. The maximum curvatures of the first bending, which can be controlled by the deflection (see details in Supplementary Text 5), are applied as $0.4\kappa_e$, κ_e , $1.4\kappa_e$, $2\kappa_e$, $2.4\kappa_e$, and $2.9\kappa_e$, as shown in Fig. 3 (C to H), respectively.

1) For the maximum curvature of the first bending in $0 \leq \kappa_1 \leq \kappa_e$ (Fig. 3, C and D), the force-deflection curves of the four steps all

overlap, indicating that plastic behavior does not occur during the loading-unloading cycles. The elastic bendability is not enhanced.

2) For the maximum curvature of the first bending in $\kappa_e < \kappa_1 \leq 2\kappa_e$ (Fig. 3, E and F), the force-deflection curve of first reverse unloading does not overlap with the force-deflection curve of the first bending, suggesting that plastic behavior occurs during the first bending. However, the force-deflection curves of the first reverse unloading, second bending, and second reverse unloading all overlap, indicating that the stack undergoes pure elastic behavior without any plastic behavior during the three steps. Therefore, the enhanced elastic bendability is identical to the maximum curvature of the first bending.

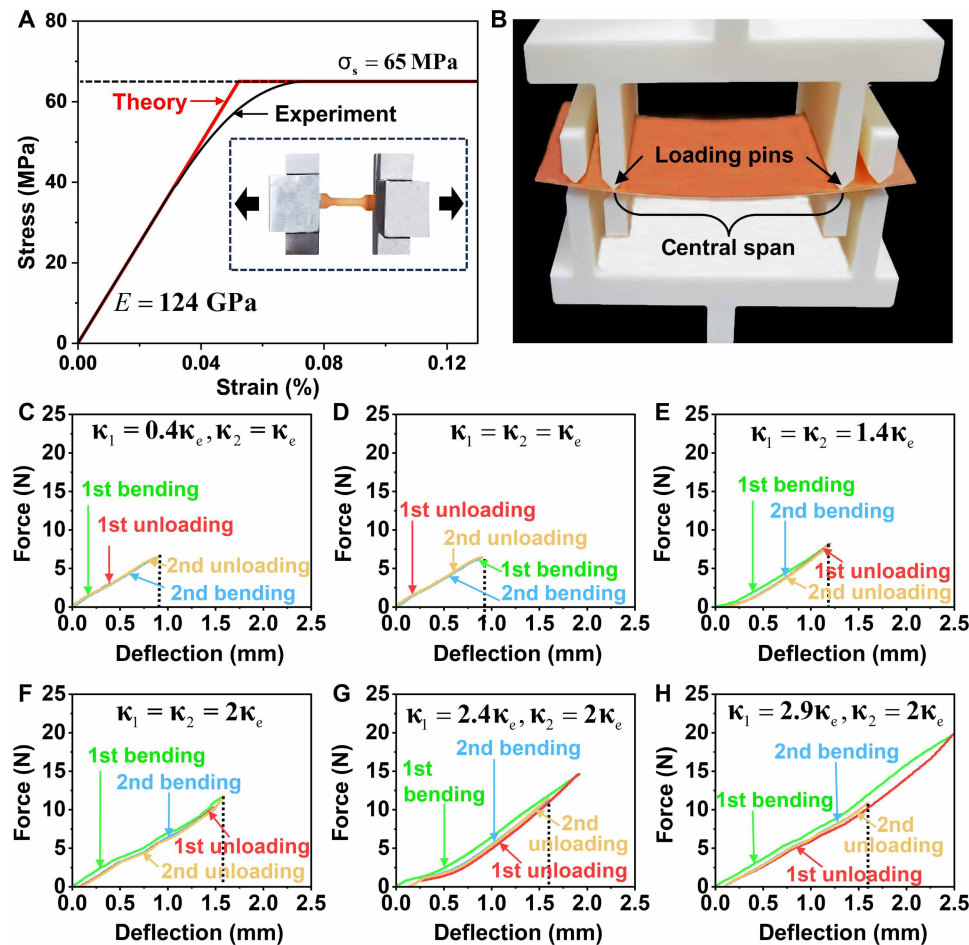


Fig. 3. Experimental verification of the overbend strategy. (A) Stress-strain curves of a dog bone-shaped copper plate in the tensile test (10 times). The subfigure is the tensile test diagram. (B) Four-point bending experimental diagram. (C to H) Force-deflection curves for different overbend values, consisting of four steps: the first bending, the first reverse unloading, the second bending, and the second reverse unloading.

3) For the maximum curvature of the first bending in $\kappa_1 > 2\kappa_e$ (Fig. 3, G and H), the curves of the first bending, first reverse unloading, and second bending do not overlap, indicating that plastic behavior occurs during the first bending and first reverse unloading. When the second bending is applied in $0 < \kappa_2 \leq 2\kappa_e$, the curves of the second bending and second unloading overlap, suggesting that no plastic behavior occurs in this range. However, when the second bending is applied in $\kappa_2 > 2\kappa_e$, the curves of the second bending and second unloading do not overlap, indicating that plastic behavior occurs again (fig. S8). These results confirm that the maximum enhanced elastic bendability is $2\kappa_e$, which is also consistent with the FEA and theoretical results (Fig. 2B).

The overbend strategy is applied to the flexible curvature sensor to demonstrate the effectiveness (49). The flexible curvature sensor has a sandwich structure, consisting of a phenolic resin interlayer and the top and bottom Constantan foils as the sensitive grids (fig. S9), with thicknesses of $t_{\text{phenolic resin}}$ and $t_{\text{Constantan}}$, respectively. The curvature can be obtained by measuring the resistance of the top and bottom sensitive grids

$$\kappa = \frac{\Delta R_{\text{top}} - \Delta R_{\text{bottom}}}{(t_{\text{phenolic resin}} + t_{\text{Constantan}})K_s R_0} \quad (4)$$

where R_0 is the initial resistance; ΔR_{top} and ΔR_{bottom} are the changes in resistance of the top and bottom sensitive grids, respectively; and K_s is the gauge factor. Here, the effect that the elastic bendability could be doubled by the overbend strategy is verified by FEA (fig. S10). After implementing the overbend strategy for the curvature sensor, it is integrated into a glove to monitor the curvature of the finger. As shown in fig. S11, the curvature sensor maintains the elastic response even when the curvature reaches 1.6 times the designed elastic bendability, demonstrating that the strategy could take effect in practice (see details in Supplementary Text 6).

Effects of geometric and material parameters on the enhanced elastic bendability

The effects of geometric and material parameters on the enhanced elastic bendability are analyzed in this section. Figure 4 (A and B) shows the designed elastic bendability and maximum enhanced elastic bendability after the execution of the overbend strategy for various thicknesses of the metal layer (ideal elastoplastic constitutive relationship) and polymer layer. As noted earlier, the designed elastic bendability decreases with increasing thicknesses of the metal/polymer layers. When the thickness of the metal layer is reduced from 100 to 5 μm , the designed elastic bendability increases from

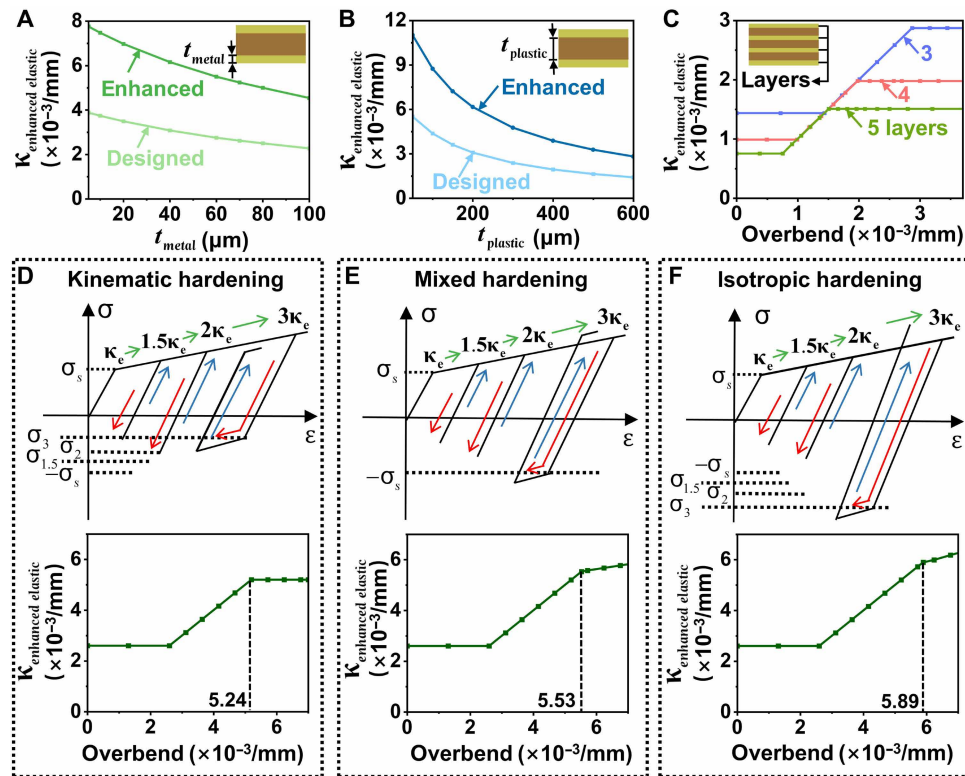


Fig. 4. Effects of geometric and material parameters of the overbend strategy. (A) Designed elastic bendability and maximum enhanced elastic bendability for different metal thicknesses. (B) Designed elastic bendability and maximum enhanced elastic bendability for different polymer thicknesses. (C) Enhanced elastic bendability for various layers of metal with different overbend values. (D to F) Evolution of different constitutive relations and enhanced elastic bendability for various plastic hardening with different overbend values.

$2.27 \times 10^{-3}/\text{mm}$ to $3.88 \times 10^{-3}/\text{mm}$. Similarly, when the thickness of the polymer layer is reduced from 600 to 50 μm , the designed elastic bendability increases from $1.14 \times 10^{-3}/\text{mm}$ to $5.51 \times 10^{-3}/\text{mm}$. For various thicknesses of the metal and polymer layers, the overbend strategy can effectively double the designed elastic bendability. Figure 4C shows the designed elastic bendability and enhanced elastic bendability of multilayer stacks with three or more metal layers, which demonstrate that the strategy can double the elastic bendability for multilayer flexible electronics. It can be inferred that the strategy also takes effect for single-layer flexible electronics.

In addition, plastic hardening is considered for certain materials. Figure 4 (D to F) presents the effects of the overbend strategy for the linear hardening constitutive relationship with hardening modulus $E_{\text{hardening}} = 12.4 \text{ GPa}$ under different hardening models, including kinematic hardening, mixed hardening (the weight ratio between kinematic and isotropic hardening is equal here), and isotropic hardening, respectively. For the kinematic hardening constitutive relationship, because of the decrease in the reverse yield stress with the increase in the forward yield stress, the stress range maintains a constant value $2\sigma_s$. Thus, the maximum enhanced elastic bendability remains twice the designed elastic bendability. For the mixed hardening constitutive relationship, the maximum enhanced elastic bendability can exceed twice the designed elastic bendability. The forward yield stress increases, while the reverse yield stress remains constant, leading to an expanded elastic stress range. For the isotropic hardening constitutive relationship, the maximum enhanced elastic bendability can also be more than twice the elastic

bendability, with a superior enhanced effect. The reason is that the expansion rate of the elastic stress range is greater than that in the mixed hardening because of the increase in reverse yield stress with the increase in the forward yield stress. For instance, after overbending to $7.02 \times 10^{-3}/\text{mm}$, the enhanced bendability reaches $5.2 \times 10^{-3}/\text{mm}$ for the kinematic hardening constitutive relationship, $5.61 \times 10^{-3}/\text{mm}$ for the mixed hardening constitutive relationship, and $6.4 \times 10^{-3}/\text{mm}$ for the isotropic hardening constitutive relationship. The horizontal coordinate of the dashed line is the critical value. Between the critical value and the designed elastic bendability, the enhanced elastic bendability is exactly equal to the curvature of the overbend. The critical value is the largest for the isotropic hardening relationship and the smallest for the kinematic hardening relationship. In addition, the effects of the ratio between the kinematic hardening and the isotropic hardening are studied of 2:8, 4:6, 6:4, and 8:2 (fig. S12). The critical value increases as the proportion of isotropic hardening increases. In practice, aluminum and copper are close to the kinematic hardening model, and platinum and titanium are close to the isotropic plastic hardening model, respectively, while the ratio between the proportions of the kinematic hardening and isotropic hardening is about 1:1 for silver. When the maximum curvature of the first bending exceeds the critical value, the enhanced elastic bendability increases as the proportion of isotropic hardening increases. Thus, considering the hardening effects of metal, the isotropic hardening model is more advantageous for achieving a higher elastic bendability via the overbend strategy. The effects of the hardening modulus are investigated

for values of 0.1, 0.5, 1, and 5 GPa (fig. S13). For the maximum curvature in the first bending below twice the designed elastic bendability, the effects of the overbend strategy are independent of the hardening modulus, while both the critical value and enhanced elastic bendability increase with the increase in hardening modulus for the maximum curvature in the first bending beyond twice the designed elastic bendability.

Considering that many flexible electronics are subjected to bending in two opposite directions, the performance can also be optimized by using the overbend strategy. For the material with the ideal elastoplastic/kinematic hardening constitutive relationship, the elastic bendability in two directions can be balanced through the strategy, which optimally utilizes the elastic properties of the material. For instance, the multilayer stack can be overbent inward to 1.5 times the designed elastic bendability and then returned to the original state. In this case, the elastic bendability in the inward direction is 1.5 times the designed elastic bendability, while the elastic bendability in the outward direction is 0.5 times the designed elastic bendability (fig. S11). For the material with the mixed hardening/isotropic hardening constitutive relationship, the overbend strategy can be used to enhance the elastic bendability in two directions simultaneously (see details in Supplement Text 7 and fig. S14).

The fatigue of bendable structures, which is related to the deformation amplitude and elastoplastic behaviors, should be considered for the design of flexible electronics. The requirement of the fatigue life relies on the application purpose and service environment (see details in Supplement Text 8). For instance, the short-term fatigue life is enough for the biodegradable implantable microelectrode arrays, which could detect and treat cardiac complications (50), followed by complete bioresorption via hydrolysis within 6 weeks. In contrast, flexible sensors integrated into aircrafts need to continuously monitor the deformation for several decades, requiring a long-term fatigue life. Here, the effects of the overbend strategy on fatigue are studied because the plastic behavior occurs during the operation. Supposing that the maximum curvature of the cyclic bending remains below the designed elastic bendability, the fatigue life is relatively long because only the elastic behavior occurs. When the maximum curvature of the first bending lies between the designed elastic bendability and the critical value and the maximum curvature of the subsequent cycles is below that of the first bending, plastic behavior occurs only in the first cycle and only purely elastic behavior occurs in the subsequent cycles (51–54). In this case, the bendable structure is in a shakedown state and the fatigue behavior is close to that in the elastic state. When the maximum curvature of each bending exceeds the critical value, plastic behavior occurs in each bending cycle, which may lead to the low-cycle fatigue for the flexible electronics (55, 56). In summary, the analysis of the overbend strategy provides the insight for the fatigue design: Long fatigue life could be maintained not only within the designed elastic range but also within twice or more times that range.

Mechanical analysis of interconnects bonded on the soft substrate

In practical applications of flexible electronics, interconnects with various geometric designs bonded on the soft substrate are frequently used to provide exceptional elastic stretchability (57–68). The ideal elastoplastic constitutive relationship is adopted to the metal for

simplicity. The appearance of the plastic behavior is defined as the limit of elastic bendability. Figure 5A shows the bending of the serpentine interconnect bonded on the soft substrate. It is observed by the FEA that the designed elastic bendability increases as the thickness increases from 0.01 to 0.6 μm and then decreases as the thickness increases from 0.6 to 100 μm (Fig. 5B). For the thicknesses between 0.01 and 0.6 μm , the serpentine interconnect experiences buckling/wrinkling deformations at the arc tops. The wrinkles become more densely packed with the thickness decreases, as shown in the left subfigure of Fig. 5B. For the thicknesses between 0.6 and 100 μm , the buckling/wrinkling effect is diminished or even absent, as shown in the right subfigure of Fig. 5B. The designed elastic bendability decreases with the increase in the thickness because of the higher stress for larger thickness.

Figure 5 (C to E) show the enhanced elastic bendability of serpentine, zigzag, and fractal interconnects (fig. S15) with the thickness of 5 μm bonded on the soft substrate, the deformation of which is without buckling. The regularity and conclusion similar to the above can be obtained. For the three types of interconnects, respectively, if the first bending does not exceed the designed elastic bendability $\kappa_{\text{overbend}} \leq 0.41 / \text{mm} / \kappa_{\text{overbend}} \leq 0.255 / \text{mm} / \kappa_{\text{overbend}} \leq 0.30 / \text{mm}$, the elastic bendability is not enhanced, i.e., $\kappa_{\text{enhanced elastic}} = \kappa_{\text{designed elastic}}$. If the first bending is between one and two times the designed elastic bendability $0.41 / \text{mm} < \kappa_{\text{overbend}} \leq 0.82 / \text{mm} / 0.255 / \text{mm} < \kappa_{\text{overbend}} \leq 0.51 / \text{mm} / 0.30 / \text{mm} < \kappa_{\text{overbend}} \leq 0.60 / \text{mm}$, the enhanced elastic bendability reaches the maximum value of the first bending, i.e., $\kappa_{\text{enhanced elastic}} = \kappa_{\text{overbend}}$. If the first bending exceeds twice the designed elastic bendability $\kappa_{\text{overbend}} \geq 0.82 / \text{mm} / \kappa_{\text{overbend}} \geq 0.51 / \text{mm} / \kappa_{\text{overbend}} \geq 0.60 / \text{mm}$, the enhanced elastic bendability remains twice the designed elastic bendability, i.e., $\kappa_{\text{enhanced elastic}} = 0.82 / \text{mm} / \kappa_{\text{enhanced elastic}} = 0.51 / \text{mm} / \kappa_{\text{enhanced elastic}} = 0.60 / \text{mm}$. We also add microscopic images indirectly demonstrating the effect of the strategy (see details in Supplement Text 9). These results demonstrate that the overbend strategy is effective not only for simple straight single-layer foils and multilayer stacks but also for interconnects with various geometric designs. More complicated structures consisting of multiple layers of interconnects embedded within the soft substrate, each of which has different geometric designs, are also studied. It confirms the findings that the overbend strategy can enhance the elastic bendability to twice or more times for thick and thin interconnects (fig. S16).

In addition, the interconnects are often subjected to stretching-bending coupling deformation when the entire structure is bonded on the other thick substrate. For example, the interconnect experiences both stretching and bending when bonded to the human joint skin as the underlying substrate bends (Fig. 5F). Figure 5G shows that the designed elastic bendability first increases and then decreases with the increase in the thickness of the metal layer, which is consistent with the above results. Two representative thicknesses and deformation modes are selected to investigate the effect of the overbend operation. For $t_{\text{metal}} = 0.05 \mu\text{m}$ (Fig. 5H), the buckling mode appears. The designed elastic bendability is 0.013/mm, and the maximum enhanced elastic bendability exceeds twice the designed elastic bendability. The mechanism can be understood in the following. When the first bending exceeds twice the designed elastic bendability, the first reverse unloading leads to out-of-plane buckling deformation because of the compression, which results in purely elastic behavior occurring in the first reverse unloading. Thus, the second bending process with the same maximum value is also purely elastic, and the enhanced elastic bendability can exceed twice

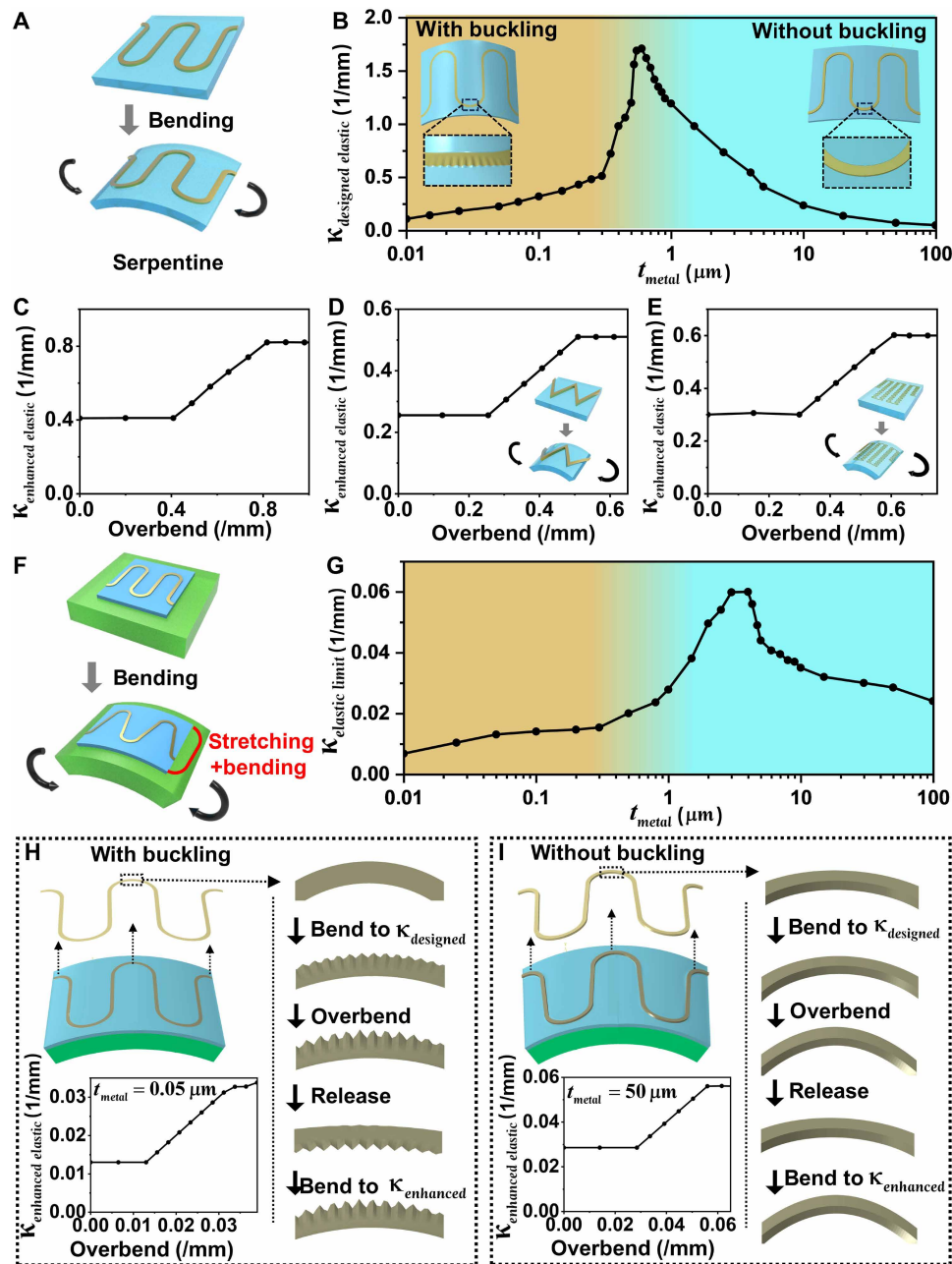


Fig. 5. Mechanical analysis of the interconnects bonded on the soft substrate. (A) Bending of the serpentine interconnect bonded on the soft substrate. (B) Designed elastic bendability of the serpentine interconnect with various thicknesses. (C to E) Enhanced elastic bendability of serpentine/zigzag/fractal interconnects bonded on the soft substrate as a function of the maximum of the first bending. (F) Stretching-bending coupling deformation of the serpentine interconnect. (G) Designed elastic bendability of the serpentine interconnect in the stretching-bending coupling deformation with various thicknesses. (H and I) Different deformation models and enhanced elastic bendability of the different metal thicknesses: $t_{\text{metal}} = 0.05 \mu\text{m}$ and $t_{\text{metal}} = 50 \mu\text{m}$.

the designed elastic bendability, as shown in the subfigure of Fig. 5H. The equivalent plastic strain and principal strain at the arc are shown in fig. S17. For $t_{\text{metal}} = 50 \mu\text{m}$ (Fig. 5I), the deformation is without buckling. The designed elastic bendability is 0.028/mm, and the maximum enhanced elastic extensibility only reaches twice the designed elastic bendability $\kappa_{\text{enhanced elastic}} = 0.056/\text{mm}$. That is because no buckling occurs during the loading-unloading cycles, the plastic behavior occurs in the first reverse unloading when the

first bending exceeds twice the designed elastic bendability. Consequently, the enhanced elastic bendability does not exceed twice the designed elastic bendability.

DISCUSSION

In this study, we clearly describe the core mechanisms and effects of the overbend strategy by using the elastoplastic bending of the

multilayer stack as an example. Through the operation beyond the designed elastic range of the structure, the elastic strain range at the critical section can be extended to twice or more times the original value. Taking multilayer stacks and interconnects with various geometric designs as examples, it is verified that the strategy can effectively enhance the designed elastic bendability to twice for the material of the ideal elastoplastic/kinematic hardening constitutive relationship and to more than twice for the mixed hardening/isotropic hardening constitutive relationship. For practical flexible electronics subjected to twisting, stretching, and coupling modes, the overbend strategy can be extended to overtwist, overstretch, and general overload strategies. It can be inferred that the enhancement of the elastic range also takes effect (see details in Supplement Text 10). The strategy, which is costless and easy to operate, may become a commonly used method to enhance the elastic range of flexible electronics.

MATERIALS AND METHODS

Preparation of the multilayer stack

The experimental multilayer stack is fabricated by applying upper and lower layers of copper foil with $t_{\text{metal}} = 70 \mu\text{m}$ and a glass resin layer with $t_{\text{polymer}} = 260 \mu\text{m}$ and $L \times w = 100 \text{ mm} \times 50 \text{ mm}$ of the entire stack (fig. S7).

Copper metal constitutive model mechanical testing

A picosecond laser precision machining system (DCT-DL566PU, China; as shown in fig. S18) is used to cut 70- μm -thick copper plates and prepare samples. Tensile testing of the copper plate is performed using an electronic material-testing machine (ZQ-990LB, China; fig. S19).

Four-point bending experiment

Relevant fixtures (fig. S6) are printed using a three-dimensional printer (Bambu Lab P1S, China; fig. S20), and the bending experiment of the multilayer stack (fig. S21) is performed using the electronic material-testing machine (ZQ-990LB, China; fig. S19). A displacement control loading mode is adopted. To demonstrate all cases of the overbend strategy, the maximum values of the first bending are chosen as $0.4\kappa_e$, κ_e , $1.4\kappa_e$, $2\kappa_e$, $2.4\kappa_e$, and $2.9\kappa_e$; the second bending of each case is shown in Fig. 3 (C to H).

Finite element analysis

Commercial software ABAQUS is used for the FEA. A half-period model was used, with boundary conditions defined by rotational displacement. The metal, glass resin core, and soft substrate are modeled using C3D8 elements. When the interconnection thickness is larger than $1 \mu\text{m}$, C3D8 solid elements are used for the interconnect. When the interconnection thickness is less than $1 \mu\text{m}$, S4R shell elements are used for the interconnect. The mesh density here already satisfies the convergence requirement.

The Ecoflex is chosen as the substrate material, which has excellent elastic stretchability and is easy to process. It is regarded as a hyperelastic material described by the Mooney-Rivlin model, and the material parameters are $C_{10} = 0.008054 \text{ MPa}$, $C_{01} = 0.002013 \text{ MPa}$, and $D_1 = 2 \text{ MPa}^{-1}$. The lowest substrate is polydimethylsiloxane, which is commonly used in microelectronic packaging. Young's modulus and Poisson's ratio are 1 MPa and 0.48 , respectively. For the flexible curvature sensor, Young's moduli and Poisson's ratios are 4 GPa and

0.35 for phenolic resin (interlayer) and 163 GPa and 0.33 for Constantan (sensitive grids), respectively.

Supplementary Materials

The PDF file includes:

Supplementary Text

Figs. S1 to S24

Legends for movies S1 to S4

Other Supplementary Material for this manuscript includes the following:

Movies S1 to S4

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An overbend strategy to manifold enhance the designed elastic bendability of flexible electronics

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